

Tuples [read only datatypes, cannot change the datatype once it is executed]

fruits=('apple','orange')

```
In [ ]: s="IoT"

s=list(s)
#tuple(s)

print(s)
print(type(s))
```

```
['I', 'o', 'T']
<class 'list'>
```

t=(1,2,3,4,2) print(t.index(3)) --> value on 3 print(t.count(2)) --> no. of occurrences of 2

```
In [6]: l=[1,2,3,4]

for x in l:
    print(x)
```

```
1
2
3
4
```

```
In [ ]: t=(1,2,3,4,5)          tuple
print(t.index(3))
print(t.count(2))
```

```
2
1
```

subjects from user, empty list add in list

```
In [17]: subject=[]

for x in range(4):
    a = input("Enter subject name")
    subject.append(a)

print(subject)
```

```
['Math', 'Operating System', 'IoT', 'Machine Learning']
```

dictionary

hash table key value pair

```
In [7]: student={'name':'ali','roll no':'6006','major':'EE'}
```

```
print(type(student))
print(len(student))
print(student['name'])
print(student.items())
print(student.keys())
print(student.values())
```

```
print('\n')
stud={'1':student, '2':'class'}
print(stud)
```

1 & 2 is key

```
<class 'dict'>
```

```
3
```

```
ali
```

```
dict_items([('name', 'ali'), ('roll no', '6006'), ('major', 'EE')])
```

```
dict_keys(['name', 'roll no', 'major'])
```

```
dict_values(['ali', '6006', 'EE'])
```

```
{'1': {'name': 'ali', 'roll no': '6006', 'major': 'EE'}, '2': 'class'}
```